

Mutual Fund Analytics Platform

Day 1 – Data Ingestion and Validation

Internship Report submitted to
Bluestock Fintech

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Day 1 deliverable – project foundation, data ingestion, and validation.

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Abstract

This report covers Day 1 activities of the Mutual Fund Analytics Platform project at Bluestock Fintech. The work involved setting up the project environment, loading and inspecting ten mutual fund datasets, exploring fund master metadata, validating AMFI codes, and integrating live NAV data from the MFAPI service. All datasets were found to be in good condition, AMFI code validation passed with zero missing matches, and live NAV data was successfully fetched for six large-cap schemes.

Project Overview

Organization: Bluestock Fintech

Date: 23/06/2026

Objective: Build the data ingestion and validation foundation for a Mutual Fund Analytics Platform by organizing datasets, understanding fund metadata, validating AMFI codes, and integrating live NAV data.

Work Completed

- Created project repository and connected it to GitHub
- Set up folder structure with separate directories for raw data, processed data, notebooks, SQL scripts, dashboard, and reports
- Installed all required dependencies and generated `requirements.txt`
- Loaded and inspected ten mutual fund datasets
- Explored and analyzed fund master metadata
- Validated AMFI codes between fund master and NAV history
- Fetched live NAV data for six large-cap schemes via MFAPI

Project Folder Structure

Folder / File	Purpose
data/raw/	Original source datasets (CSV files)
data/processed/	Cleaned and transformed datasets
notebooks/	Jupyter notebooks for exploration
sql/	SQL scripts for querying
dashboard/	Dashboard and visualizations
reports/	Generated reports
data_ingestion.py	Loads and inspects all datasets
explore_fund_master.py	Explores fund master metadata
fetch_nav_5.py	Fetches NAV for five selected schemes
live_nav_fetch.py	Fetches live NAV from MFAPI
validate_amfi.py	Validates AMFI codes across datasets
requirements.txt	Python package dependencies
README.md	Project overview and setup guide
.gitignore	Git ignore configuration

Dataset Overview

No.	Dataset	Records
1	Fund Master	40 rows, 15 columns
2	NAV History	46,000 rows, 3 columns
3	AUM by Fund House	90 rows, 5 columns
4	Monthly SIP Inflows	48 rows, 6 columns
5	Category Inflows	144 rows, 3 columns
6	Industry Folio Count	21 rows, 6 columns
7	Scheme Performance	40 rows, 19 columns
8	Investor Transactions	32,778 rows, 13 columns
9	Portfolio Holdings	322 rows, 8 columns
10	Benchmark Indices	8,050 rows, 3 columns

Data Ingestion Summary

All ten datasets were loaded using Pandas. The following observations were made:

- Total datasets analyzed: **10**
- All files loaded successfully without import errors
- Data types were appropriate for analytical use in all datasets
- Most datasets were complete with **no missing values**
- **Monthly SIP Inflows** had missing values in the Year-on-Year growth percentage field only
- No major structural issues or duplicate records were identified
- Date columns were converted to proper datetime format where required

Fund Master Exploration

Summary

Attribute	Value
Total Schemes	40
Total Fund Houses	10
Categories	Equity, Debt
Sub-Categories	12
Risk Profiles	5

Fund Houses (Examples)

- SBI Mutual Fund
- HDFC Mutual Fund
- ICICI Prudential Mutual Fund
- Nippon India Mutual Fund
- Kotak Mahindra Mutual Fund

Sub-Categories (Examples)

- Large Cap, Mid Cap, Small Cap, Flexi Cap
- ELSS (Tax Saving), Index Funds
- Debt-oriented categories

Understanding AMFI Codes

- Each mutual fund scheme registered in India is assigned a unique numeric code by AMFI (Association of Mutual Funds in India).
- The AMFI code is the primary key in the Fund Master dataset.
- It links fund metadata in the Fund Master to daily NAV records in the NAV History dataset.
- It is also used by the MFAPI service to fetch live and historical NAV data.
- Consistent AMFI codes across datasets ensure reliable joins and analysis.

Live NAV Data Integration

Live NAV data was fetched from **MFAPI** (api.mfapi.in), a free public REST API for AMFI-registered schemes. NAV responses were successfully retrieved for all six schemes and stored as CSV files inside the raw data directory.

AMFI Code	Scheme Name	Fetch Status
125497	HDFC Top 100 Direct	Success
119551	SBI Bluechip	Success
120503	ICICI Bluechip	Success
118632	Nippon Large Cap	Success
119092	Axis Bluechip	Success
120841	Kotak Bluechip	Success

AMFI Validation Results

AMFI codes were cross-referenced between the Fund Master and NAV History datasets using `validate_amfi.py`.

Validation Check	Result
Total AMFI codes in Fund Master	40
Total AMFI codes in NAV History	40
Missing codes (in either dataset)	0
Linkage integrity	Fully maintained

- Every scheme in the Fund Master has corresponding NAV history records.
- No orphan codes were found in either direction.
- The datasets are fully aligned and ready for joins and analysis.

Data Quality Assessment

Dataset	Missing Values	Overall Quality
Fund Master	None	Good
NAV History	None	Good
AUM by Fund House	None	Good
Monthly SIP Inflows	YoY field	Acceptable
Category Inflows	None	Good
Industry Folio Count	None	Good
Scheme Performance	None	Good
Investor Transactions	None	Good
Portfolio Holdings	None	Good
Benchmark Indices	None	Good

Key Findings

- All ten datasets loaded and inspected successfully.
- Fund master covers 40 schemes across 10 fund houses and 2 broad categories.
- AMFI code validation passed with a perfect match – zero discrepancies.
- Live NAV data was fetched and stored for all six target large-cap schemes.
- Only one field (YoY growth in SIP inflows) contained missing values.
- No duplicate records were found across any dataset.
- Data types and date formats are consistent and analysis-ready.
- The project foundation is complete and ready for Day 2 analytics work.

Conclusion

Day 1 of the Mutual Fund Analytics Platform project has been completed successfully. The project environment is set up, all datasets are loaded and validated, AMFI code integrity is confirmed, and live NAV data has been integrated. The datasets are clean, consistent, and ready for deeper analysis in the next phase.